AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

Chemistry Paper II

Time: 2 hours 20 minutes Marks: 55

AKUEB May 2018
Teaching & Learning Only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

a. What is the trend of acidity of the oxides of group IVA elements? (1 Mark)

b. Compare the reactivity of carbon dioxide (CO_2) and silicon dioxide (SiO_2) with respect to their reaction with

i. water. (2 Marks)

ii. sodium hydroxide. (2 Marks)

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Q.2. (Total 5 Marks)

a. Write a complete balanced chemical equation to show the reaction of $KMnO_4$ with

i. iron sulphate ($FeSO_4$). (2 Marks)

ii. oxalic acid ($H_2C_2O_4$). (2 Marks)

b. Identify the substance which is reduced in the reaction in part i. (1 Mark)

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Q.3. (Total 4 Marks)

a. Why are Grignard reagents more reactive as compared to most of the other organic compounds? (1 Mark)

_____

_____

b. Write a chemical equation to show the reaction of

i. propyl magnesium bromide with carbon dioxide. (1 Mark)

_____

_____

ii. methyl magnesium bromide with propanone. (1 Mark)

_____

_____

iii. ethyl magnesium bromide with ethanal. (1 Mark)

_____

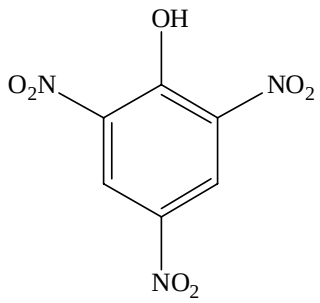
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Q.4. (Total 5 Marks)

a. Write the IUPAC names of the following phenols.

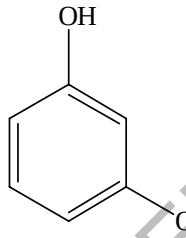
i. (1 Mark)



_____

_____

ii. (1 Mark)

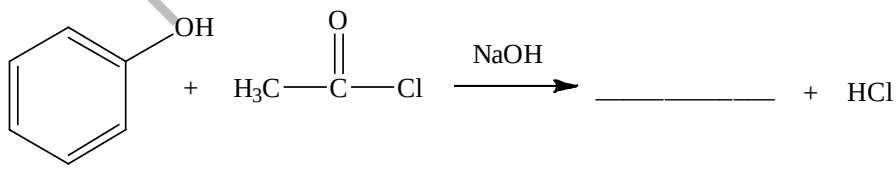


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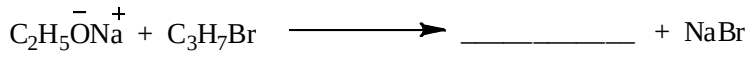
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b. Complete the given reactions by writing the missing products. (3 Marks)

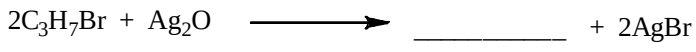
i.



ii.



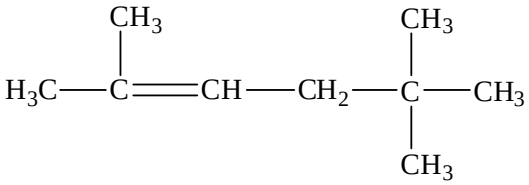
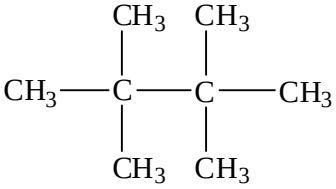
iii.



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Q.5. (Total 4 Marks)

a. Write the IUPAC names of the given organic compounds following the rules of nomenclature. (2 Marks)

Organic Compound	IUPAC Name
	
	

b. Write a chemical equation to show the reaction of 1-butyne with ozone (O_3). (2 Marks)

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Q.6. (Total 4 Marks)

a. Which of the following carboxylic acids is more acidic? Give reason for your selection. (2 Marks)

- FCH_2COOH
- $ClCH_2COOH$

b. What will happen if ethanoic acid reacts with sodium metal? (2 Marks)

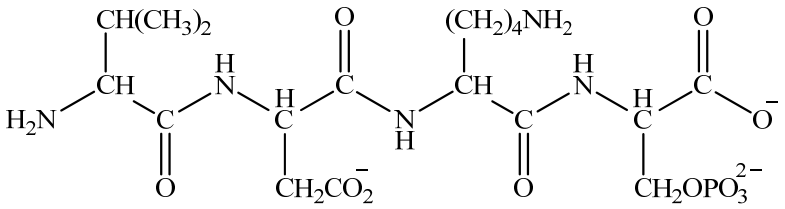
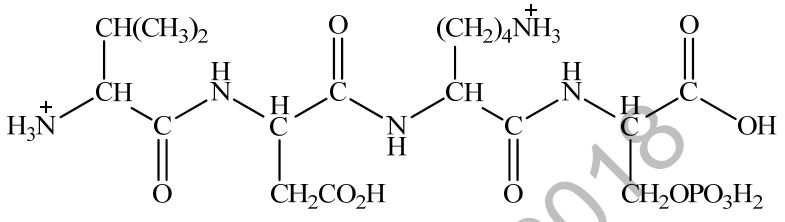
Q.7. (Total 4 Marks)

Name and describe any TWO sources of organic compounds.

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Q.8. (Total 5 Marks)

Casein, a milk protein, contains a large number of phosphate groups attached to the amino acid side chains in its polypeptide structure. The effect of pH on the solubility of casein reflects the ionisation of the acidic and basic groups in its structure.

X	
Y	

a. Identify the structure (**X** or **Y**) which shows the existence of casein protein at

i. low pH. (1 Mark)

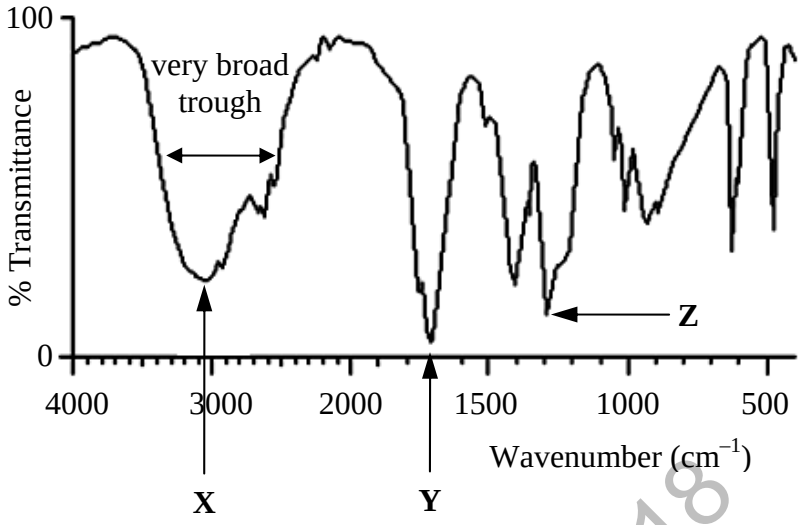
ii. high pH. (1 Mark)

b. At what pH precipitation of casein occurs? Explain your answer with reasoning. (3 Marks)

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Q.9. (Total 4 Marks)

Given below is an infrared spectrum showing peaks **X**, **Y** and **Z**.



Bond	Range (cm^{-1})
O–H (alcohols)	3230 – 3550
O–H (acids)	2500 – 3000
C=C	1620 – 1680
C=O	1680 – 1750
C–O	1000 – 1300
C–H	2850 – 3300

a. Match the peaks **X**, **Y** and **Z** with the CORRECT type of bonds in the given table. (3 Marks)

X = _____

Y = _____

Z = _____

b. Based on the information given, encircle the compound to which the spectrum belongs. (1 Mark)

i. Acetone

ii. Ethanoic acid

iii. Benzene

iv. Ethanol

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Q.10. (Total 8 Marks)

EITHER

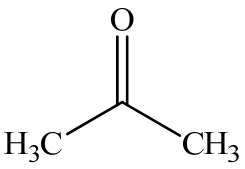
a.

i. What is resonance? (1 Mark)

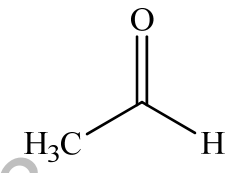
ii. Explain the resonance in benzene with the help of Kekule and Dewar structures. (7 Marks)

OR

b.



A



B

i. Which of the given carbonyl compounds (**A** or **B**) is more reactive towards nucleophile. Give a reason for your answer. (2 Marks)

ii. Write a chemical equation using carbonyl compound **A** to show the formation of (4 Marks)

I. hydrate.

II. cyanohydrin.

III. imine.

IV. oxime.

iii. Write the mechanism for the formation of cyanohydrin (formed in part II). (2 Marks)

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Q.11. (Total 7 Marks)

EITHER

a.

i. Explain any ONE environmental problem caused by the emission of each of the following.

(5 Marks)

I. Oxides of sulphur

II. Oxides of carbon

(**Note**: Write chemical equations where necessary to support your answer.)

ii. Write any TWO negative impacts on the environment of any one of the problems described above. (2 Marks)

OR

b.

i. Describe the synthesis of polyvinyl chloride (PVC) and Nylon-6,6 using chemical equations. (5 Marks)

ii. Identify the type of polymerisation involved in the synthesis of EACH polymer in part i. (2 Marks)

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